



The Ephemeris



December 2021

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December 2021 - March 2022 Events

Schedule subject to change due to Covid-19 restrictions.
Please check our website for up to date information.

Board & Speaker Talk (online)

Saturday 12/18, 1/15, 2/19, 3/19

Board Meetings: 6 -7:30pm

Speaker Talk: 8-9:30pm

Fix-It Day (2-4pm)

Currently on hold due to COVID-19 Shelter in Place

Solar Sunday (1:30-3:30pm)

Currently on hold due to COVID-19 Shelter in Place

Intro to the Night Sky Class & Houge Park 1Q In-Town

Star Party

Currently on hold due to COVID-19 Shelter in Place

RCDO Starry Nights Star Party

Currently on hold due to COVID-19 Shelter in Place

Imaging SIG Mtg

Tuesday 12/18, 1/18, 2/15, 3/15

Astro Imaging Clinics

(at Little Uvas OSP)

Currently on hold due to COVID-19 Shelter in Place

Binocular Star Gazing

Will resume in Summer 2022

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Pac-man Nebula

Photo by SJAA Member: **Francesco Meschia**

November 11– 14 2021

Location: Mountain View, CA



The Great Bear and Why Leaves Change Color in Autumn

Joe Fragola

Thousands of years ago, without the benefit of books to record information, cultural material and traditions were transmitted orally from one generation to another. Constellations and stories associated with them served as one way to explain nature and teach lessons to later generations. The constellations that we recognize today are described mainly in Greek mythology. Those star patterns are mostly god-favored heroes, animals, and objects that received a place among the stars as a memorial of their deeds. Cultures have formed different stories and pieced together different constellations from the stars, even though early sky watchers were looking at the same thing in the sky. Almost everyone is familiar with the asterism of the Big Dipper which is part of the greater constellation of Ursa Major, the Great Bear. However, for many indigenous North American people, the dipper asterism, the "bowl", represented the bear itself. They had a story that provided an explanation for why the star pattern of the bear has a long tail, as well as some observations about the season of Autumn.

The star pattern of the Bear is a very important star grouping because using two of its stars you can draw a line to find the North Star, a star which seems to stand unmoving in the North. By finding it we know the direction North and from it we find our directions. In the figure of the Bear four stars helped form his body. Some stories say that the three stars arcing away from the body formed the Bear's long and graceful tail. There is one problem - bears do not have long tails. For that, Native Americans had a story.

The story says that once, a long time ago, there lived on Earth a giant bear who was pursued by a giant tribal chief. Day after day, the chief chased the Bear, until finally one day he caught the bear by its stubby little tail.

The bear roared, but the chief held the bear tightly by its stubby tail and began to swing the bear around his head. Faster and faster went to bear until his tail was stretched all out of shape. And then the chief hurled the bear up into the sky where the bear got stuck, and he's been going around ever since. And even today the bear spins in a giant circle around the northern sky, high overhead, then down towards the horizon, with the two bright stars of his body always pointing towards the North Star. Most stories say that the four stars are a bear, but some stories say the three stars are not a tail at all. Instead, those stars are three hunters stalking the bear. The first hunter carries a

bow and arrow to shoot the bear. The last hunter carries a load of firewood to build a fire. The middle hunter carries a pot on his shoulder to cook the bear in. If you look very carefully you may be able to see the "pot", a faint star next to the middle hunter [the visual double star Alcor and Mizar].

The chase began back near the beginning of time when the first hunter shot his arrow and struck the bear in the side. The wound wasn't serious, the bear has been running ever since and the three pursuing Indians have never gained a step or have been close enough for a second shot at the bear.

Through the night, as the hunters chase the bear around the sky, the whole sky seems to be turning. But as the weeks pass, each evening the bear begins the night at a slightly different place in the sky. In the Autumn the bear begins the night low in the northwest. It is at this season every year, with the bear crouched low in the northwest that a strange thing happens. The arrow wound in the side of the bear opens slightly and a little blood trickles down upon the land. It covers the leaves of the trees and dyes them red, orange, and yellow and that is why we have Autumn.

Tribes told this story so they never forget, even when the weather is warm late into the Fall, when the bear starts his nightly journey low in the northwest it is time to store up food for the winter. The people who cannot see this wisdom in the sky and do not prepare will surely perish in the long cold nights of winter.

This is just one example of how people long ago applied observations of the night sky to their folklore in order to pass on important lessons to their descendants. The next time you're outside during a dark starry night, look up and imagine your own shapes and figures from the twinkling stars above. You'll probably end up recognizing more shapes than you would've thought, but you will also "discover" your own star patterns in the sky.

SJAA wishes its members Happy Holidays and Best Wishes for a Happy New Year 2022!



NASA's James Webb Space Telescope Primed to Lift the Haze Surrounding Sub-Neptunes

Credit: NASA

More than half of the Sun-like star systems surveyed in the Milky Way harbor a mysterious type of planet unlike any in our own solar system.

Larger than Earth, smaller than Neptune, and orbiting closer to their stars than Mercury orbits the Sun, these warm-to-hot sub-Neptunes are the most common type of planet observed in the galaxy. But although researchers have been able to measure basic properties – including size, mass, and orbit – of hundreds of these planets, their fundamental nature remains unclear.

Are they dense, Earth-like balls of rock and iron, blanketed in thick layers of hydrogen and helium gas? Or less dense mixtures of rock and ice, surrounded by steamy, water-rich atmospheres? With limited data and no planets of similar size and orbit in our own solar system to use for comparison, it has been difficult to answer these questions.

“What are these planets? How do they form? Why doesn’t our solar system have them? These are fundamental questions,” explains Jacob Bean, an astronomer at the University of Chicago who has led numerous observations of exoplanets.

The Haze Problem

The key to figuring out what sub-Neptunes are made of and how they formed is examining their atmospheres. But getting a clear view has not been easy.

The most effective method of analyzing exoplanet atmospheres is a technique known as transmission spectroscopy. When the planet is transiting its star, some wavelengths (colors) of starlight are filtered out by gases in the planet’s atmosphere. Because each type of gas has a unique “signature,” or set of wavelengths that it absorbs, it’s possible to figure out what an atmosphere is made of based on patterns in the transmission spectrum.

This technique has been successful for many exoplanets, but not for most sub-Neptunes. “There have been very few atmospheric observations of sub-Neptune planets,” explains Eliza Kempton of the University of Maryland–College Park, who specializes in theoretical modeling of exoplanet atmospheres. “And most of those have been dissatisfying in that the spectra have not revealed much in the way of spectral features that would allow us to identify the gases in the atmosphere.”

The issue seems to be aerosols, tiny particles and droplets that make up clouds or haze. These particles scatter starlight, eroding the prominent spectral peaks into subtle undulations and rendering the spectrum virtually useless in terms of determining gas composition.

But with Webb, researchers are confident that a much clearer view of sub-Neptunes is on the horizon. Two observation programs co-led by Bean and Kempton and scheduled for the first year of Webb operations will use Webb’s uniquely powerful capabilities to probe two sub-Neptune-sized planets: GJ 1214 b, the archetype sub-Neptune; and TOI-421 b, a more recent discovery.

The Sub-Neptune Archetype: GJ 1214 b

GJ 1214 b, a warm sub-Neptune orbiting a nearby red dwarf star, has been the subject of dozens of investigations. Its short orbital period, large size relative to its star, and comparative proximity to Earth make it easy (as exoplanets go) to observe effectively, while its status as the benchmark sub-Neptune – and, according to Bean, “the most mysterious exoplanet that we know of” – make it a worthy target of investigation.

The team will use Webb’s Mid-Infrared Instrument (MIRI) to stare at the GJ 1214 system nearly continuously for nearly 50 hours as the planet completes a little more than one full orbit. They will then analyze the data in three different ways to narrow down the possible combinations of gases and aerosols that make up GJ 1214 b’s atmosphere.

Transmission Spectroscopy: If molecules like water, methane, or ammonia are abundant, they should be obvious in the transmission spectrum. Mid-infrared light should not be scattered by aerosols in the same way as visible and near-infrared light.

Thermal Emission Spectroscopy: Mid-infrared light emitted by the planet itself will provide information about the planet’s temperature and reflectivity, both of which are affected by the atmosphere. A planet surrounded by dark, sooty, light-absorbing haze, for example, will be warmer than one covered in bright, reflective clouds.

Phase Curve Temperature Mapping: Although Webb will not be able to observe GJ 1214 b directly (the planet is too close to its star), it is sensitive enough to measure very subtle changes in the total amount of light from the system as the planet orbits the star. Researchers will use GJ 1214 b’s phase curve, a graph of brightness versus phase (i.e., how much of the planet’s day side is facing the telescope) to map the average temperature of the planet with longitude. This will provide additional information about the circulation and make-up of the atmosphere.

Continued on Page 4...



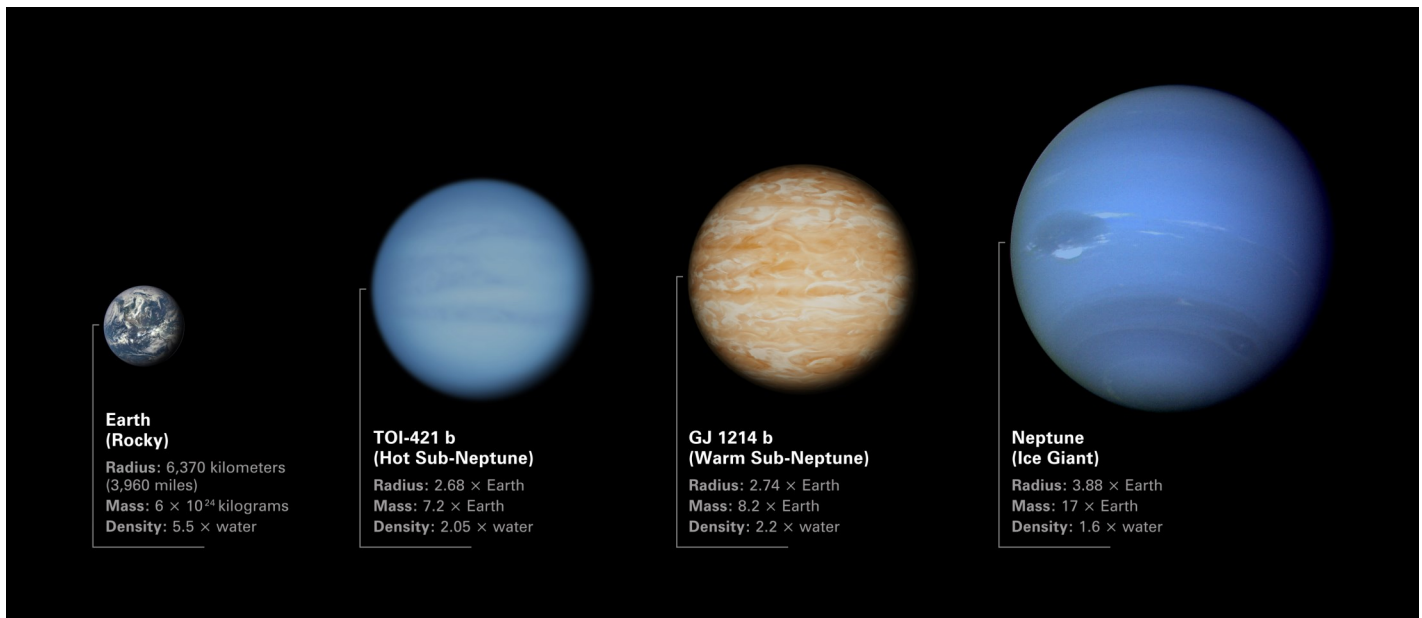


Illustration comparing the sizes of sub-Neptune exoplanets TOI-421 b and GJ 1214 b to Earth and Neptune. Both TOI-421 b and GJ 1214 b are in between Earth and Neptune in terms of radius, mass, and density. The low densities of the two exoplanets indicates that they must have thick atmospheres. The planets are arranged from left to right in order of increasing radius and mass:

Image of Earth from the Deep Space Climate Observatory: Earth is a rocky planet with an average radius of roughly 6,370 kilometers, a mass of about 6 billion trillion metric tons, and a density 5.5 times that of water.

Illustration of TOI-421 b: TOI-421 b is a hot sub-Neptune exoplanet with a radius 2.68 times Earth, a mass 7.2 times Earth, and a density 2.05 times water.

Illustration of GJ 1214 b: GJ 1214 b is a warm sub-Neptune exoplanet with a radius 2.74 times Earth, a mass 8.2 times Earth, and a density 2.2 times water.

Image of Neptune from Voyager 2: Neptune is an ice giant with a radius 3.88 times that of Earth (giving it a volume nearly 58 times Earth), a mass 17 times Earth, and a density of only 1.6 times water.

The illustration shows the planets to scale in terms of radius, but not location in space or distance from their stars. While Earth and Neptune orbit the Sun, TOI-421 b orbits a sun-like star roughly 244 light-years away, and GJ 1214 b orbits a small red dwarf star about 48 light-years away.

Credits: NASA, ESA, CSA, and D. Player (STScI)



Illustration showing what exoplanet GJ 1214 b could look like based on current information. GJ 1214 b, a warm sub-Neptune-sized exoplanet roughly 48 light-years from Earth, is one of the most studied exoplanets in the galaxy. Previous spectroscopic observations indicate that the planet is shrouded in aerosols (clouds or haze), which have thus far made it impossible to determine the composition of gases that make up its thick atmosphere.

Credits: NASA, ESA, CSA, and D. Player (STScI)

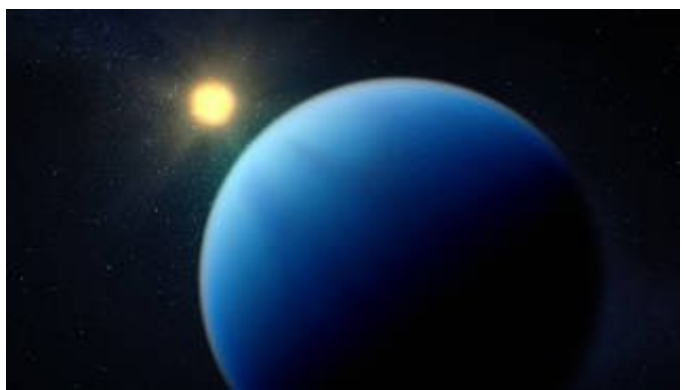


Illustration of what exoplanet TOI-421 b might look like. TOI-421 b is a hot sub-Neptune-sized exoplanet orbiting a Sun-like star roughly 244 light-years from Earth. TOI-421 b is thought to have a clear atmosphere free of haze and clouds.

Credits: NASA, ESA, CSA, and D. Player (STScI)

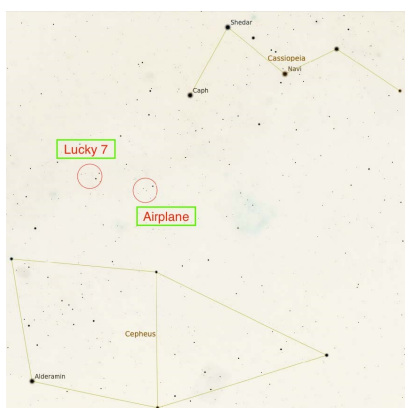


Some Fun Asterisms Recommended for Viewing

Credit: Joe Fragola

As many amateur astronomers know, the Moon spoils observations of fainter deep sky objects, such as nebulae and galaxies during part of each month. It's during these periods of time when observing is best spent focused on double stars and star clusters since these objects are less affected by moonlight. Another option includes looking for asterisms visible with telescopes or binoculars. Sometimes tracking down an interesting asterism can provide a nice alternative to viewing a new nebula or galaxy. I've included references to a few articles below that discuss observing some asterisms.

A list of asterisms was compiled by author Philip S. Harrington and the original objects on his list were called "Harrington" objects (asterisms 1 - 12). These were listed in his book "Touring the Universe Through Binoculars". As other observers contributed more objects, the list expanded, and Harrington changed the name to the STAR list (Small Telescope Asterism Roster). Later objects (STARs 12 - 29) are listed in another of his books, "The Deep Sky - An Introduction". STARs 3, 4, 9, and 12 appear in both Harrington books. A total of 29 objects are currently on the STAR list.



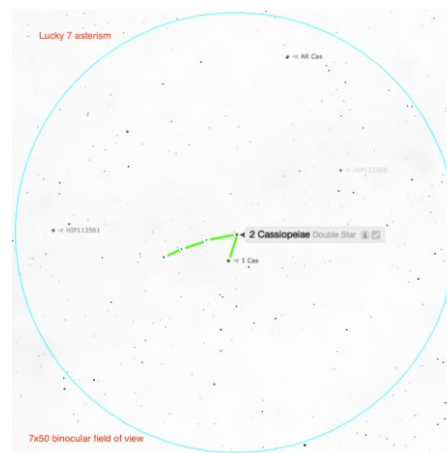
Wide field finder chart for locating two asterisms in Cassiopeia .

Credit: Joe Fragola

First up is the "Lucky 7" asterism. To find this star grouping you start at Beta Cassiopeia (Caph), the westernmost star in the "W" pattern of Cassiopeia. In 7x50 binoculars, place Beta Cas at the 1 o'clock position and scan about one and a half fields of view toward the lower left. You should see two 5th magnitude stars, 1 Cas and 2 Cas, come into view around the 7 o'clock position in the binocular field of view. These two stars form the "top" of a backward number 7.

STAR Number	Constellation	Size	RA	DEC	Name
12	Cassiopeia	60'	23h 20m	62° 20'	Airplane
29	Cassiopeia	70' X 125'	23h 03m	59° 30'	Lucky 7

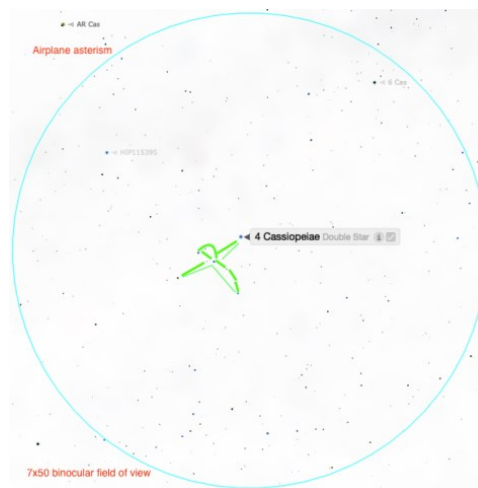
Here are the coordinates for the "Lucky 7" asterisms in case you want to make your own finder charts. STAR Number Constellation Size RA.



Binocular field of view for the "Lucky 7" asterism

Credit: Joe Fragola

The second asterism is named the "Airplane". To find it you can begin by moving the two brighter stars in the Lucky 7 asterism (1 Cas, 2 CAS) to the 9 o'clock position in your binocular field of view. Now, within this field of view you will find the 4.9 magnitude star 4 Cas. You should see the star at toward the 3 o'clock position in the field of view. This star marks the plane's eastern wing tip. A curved row of stars makes up the body of the plane and a 6th magnitude star marks the plane's western wing tip.



Binocular field of view for the "Airplane" asterism

Credit: Joe Fragola



NASA Selects Intuitive Machines for New Lunar Science Delivery

Credit: NASA



Illustration of Intuitive Machines Nova-C lander for the IM-3 mission taking four NASA investigations to Reiner Gamma

Credits: Intuitive Machines

NASA has awarded Intuitive Machines of Houston a contract to deliver research, including science investigations and a technology demonstration, to the Moon in 2024. The commercial delivery is part of NASA's Commercial Lunar Payload Services (CLPS) initiative and the Artemis program.

The investigations aboard Intuitive Machines' Nova-C lander are destined for Reiner Gamma, one of the most distinctive and enigmatic natural features on the Moon. Known as a lunar swirl, Reiner Gamma is on the western edge of the Moon, as seen from Earth, and is one of the most visible lunar swirls. Scientists continue to learn what lunar swirls are, how they form, and their relationship to the Moon's magnetic field.

"This delivery to the Moon will help the U.S. expand our capabilities and learn more about this interesting region," said Thomas Zurbuchen, associate administrator for NASA's Science Mission Directorate in Washington. "Observing lunar swirls can give us information about the Moon's radiation environment and perhaps how to mitigate its effects. With more and more science and technology demonstrations on the lunar surface, we can help prepare for sustainable astronaut missions through Artemis."

Intuitive Machines will receive \$77.5 million for the contract and is responsible for end-to-end delivery services, including payload integration, delivery from Earth to the surface of the Moon, and payload operations. This is Intuitive Machines' third task order award, the first of which is a delivery to Oceanus Procellarum on the Moon during the first quarter of 2022. This award is the seventh surface delivery task award issued to a CLPS partner.

"These investigations show how CLPS is capable of delivering payloads to the lunar surface that will address our primary scientific goals for lunar exploration and discovery," said Chris Culbert, manager of the CLPS initiative at NASA's Johnson Space Center in Houston. "We aim to

learn more about lunar swirls and this payload manifest is designed to obtain data unique to the geographical feature of Reiner Gamma."

The four investigations Intuitive Machines will deliver to Reiner Gamma are collectively expected to be about 203 pounds (92 kg) in mass and include:

Lunar Vertex is among NASA's Payloads and Research Investigations on the Surface of the Moon (PRISM) selections. It is a combination of stationary lander payloads and a rover that will make detailed measurements of the magnetic field, plasma environment and regolith properties. The lander and rover data will augment observations collected in orbit. Combined, the observations will help show how these mysterious lunar swirls form and evolve – and how they connect to local magnetic fields in the same regions. Lunar Vertex is funded through the agency's Science Mission Directorate and is led by the Johns Hopkins Applied Physics Laboratory in Laurel, Maryland.

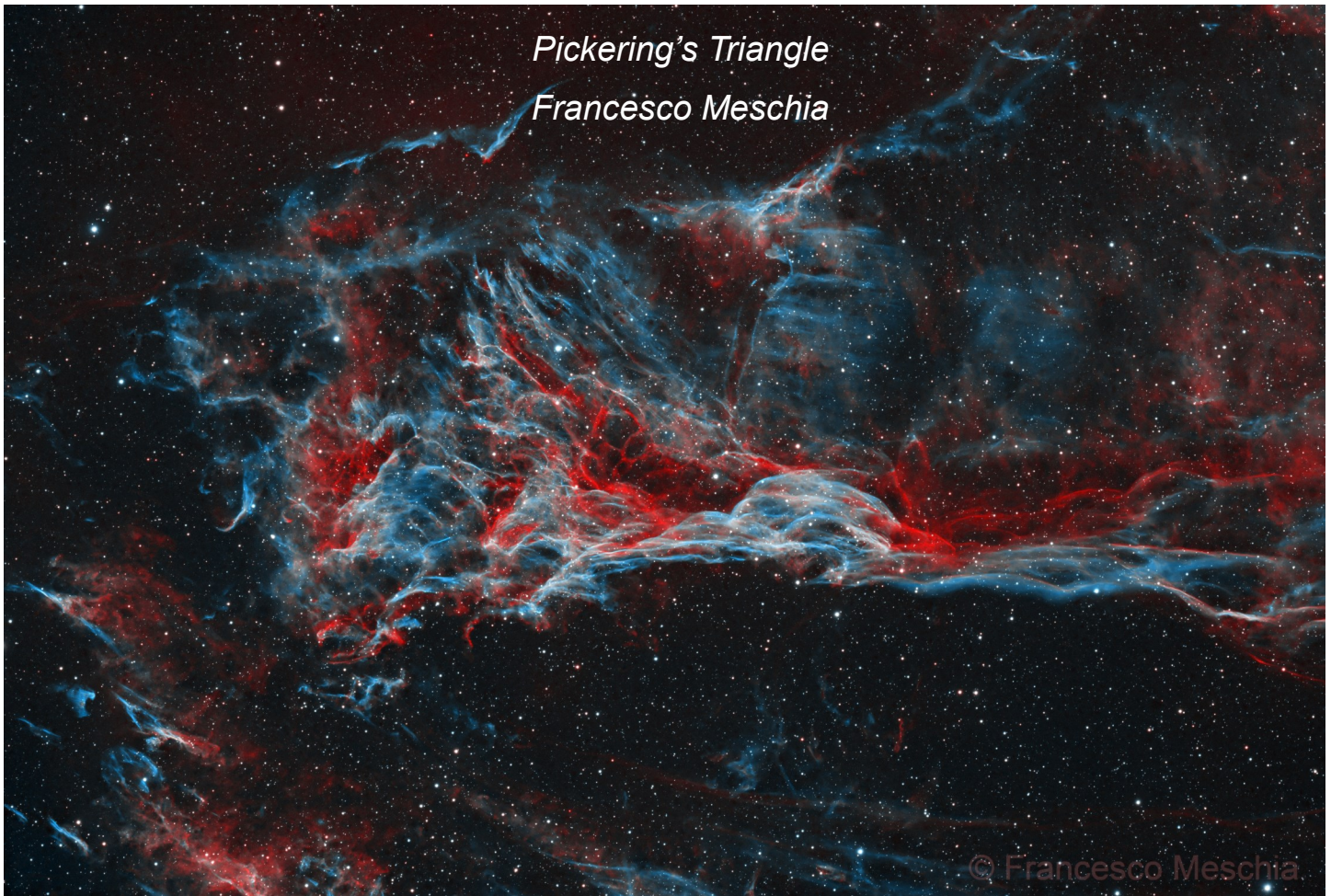
Cooperative Autonomous Distributed Robotic Exploration (CADRE) consists of mobile robots programmed to work as an autonomous team to explore the lunar surface, collect data, and map different areas of the Moon in 3D. CADRE uses its inertial measurement unit, stereo cameras, and a Sun sensor to track the position of each robot as they explore the lunar surface. CADRE is funded by NASA's Game Changing Development program under the agency's Space Technology Mission Directorate and is led by NASA's Jet Propulsion Laboratory in Pasadena, California.

MoonLIGHT retroreflector is a laser retroreflector, which reflects laser beams sent from Earth directly back from the Moon to receivers on Earth. This allows very precise measurement of the distances between the reflector and the ground station. This technique can be used to investigate relativity, the gravitational dynamics of the Earth-Moon system and the deep lunar interior. MoonLIGHT is managed by the European Space Agency (ESA).

Lunar Space Environment Monitor (LUSEM) uses a pair of apertures to detect high-energy particles on the lunar surface. LUSEM will monitor variations in the near-surface space environment when the Moon is inside and outside Earth's magnetotail – the trailing end of the magnetic fields surrounding our planet, which can serve as a buffer for incoming radiation. LUSEM is managed by the Korea Astronomy and Space Science Institute (KASI) in South Korea.

As NASA continues plans for multiple commercial deliveries to the Moon, future payloads that may be delivered with CLPS could also include other rovers, power sources, and science experiments, including technology demonstrations to later be infused into the Artemis program





Dates: Sept. 13, 2021, Sept. 14, 2021, Oct. 15, 2021, Oct. 18, 2021, Oct. 28, 2021, Nov. 4, 2021, Nov. 11, 2021

Location: Mountain View, CA

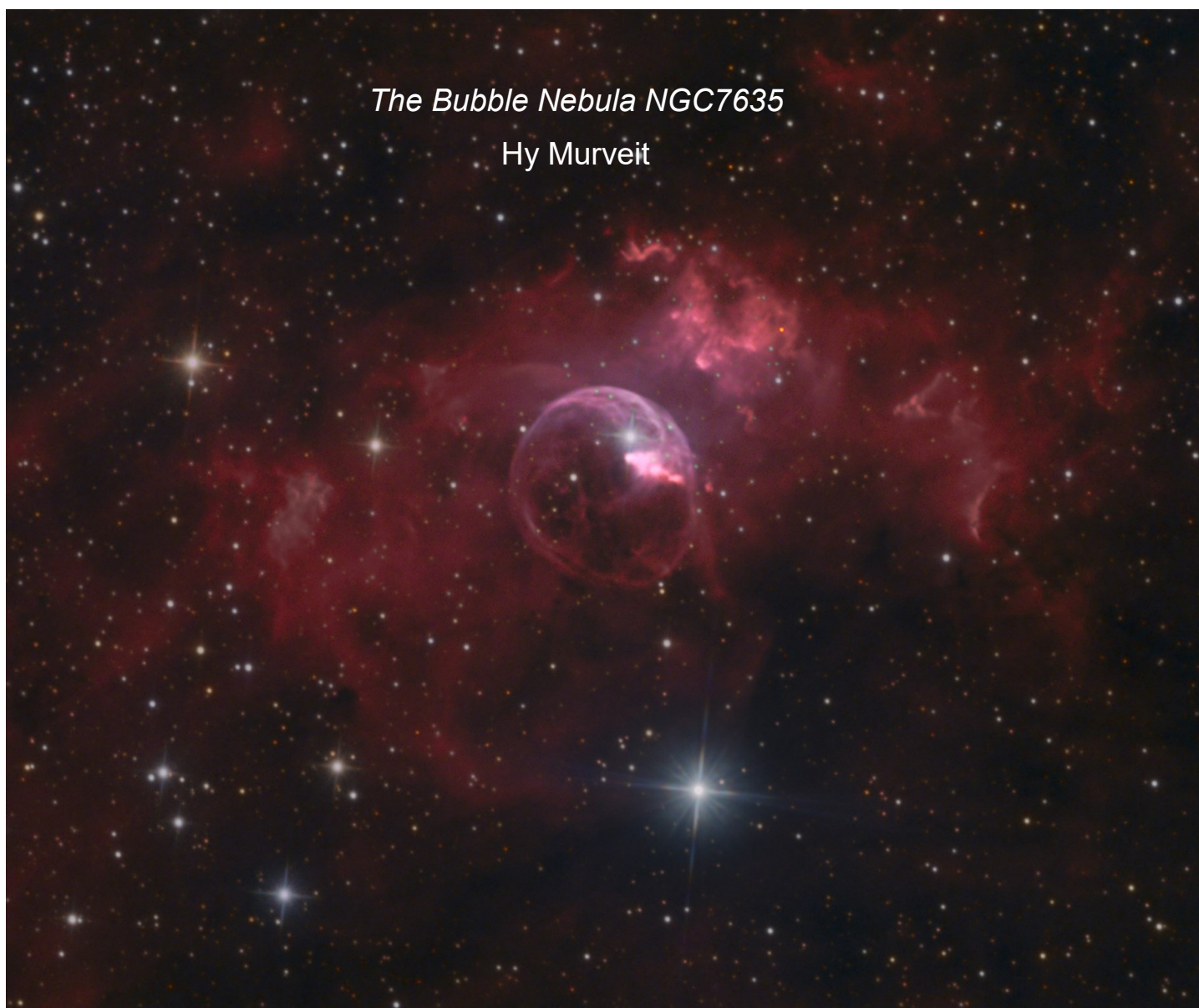
Imaging Telescope: Astro-Tech AT130EDT

Imaging Camera: ZWO ASI294MM-Pro

Mount: iOptron iEQ45 Pro

Software: N.I.N.A. Nighttime Imaging 'N' Astronomy· Pleiades Astrophoto PixInsight 1.8 Ripely





Dates: October 25, 2021

Location: Home

Imaging Telescope: GSO RC10" Truss

Imaging Camera: ZWO ASI1600mm Pro Cooled

Mount: Orion Atlas AZ/EQ Pro G

Software: Adobe Photoshop CC, KStars Ekos/INDI, Stellarmate OS, Adobe Lightroom CC Classic, PixInsight 1.8





Great Barred Spiral Galaxy NGC 1365

Nikola Nikolov

Copyright Nikola Nikolov

Date: October 16, 2021, November 7, 2021

Location: El Sauce Observatory, Rio Hurtado Valley, Coquimbo Region, Chile

Telescope: Quasar Optics 12.5" f/9 Quasar

Camera: SBIG STL11000

Mount: Losmandy Titan Gemini 2 Mini

Software: AstroPixelProcessor, Adobe Lightroom CC



SJAA Member Astrophoto Gallery



Date: June 30, 2021

Location: Chile

Telescope: PLANEWAVE CDK

Camera: FLI-PL09000

Mount: Mathis Instruments MI-1000/1250

Software: PixInsight PI 1.8, Adobe PS CC



The Road to Launch and Beyond for NASA's James Webb Space Credit: NASA

Now that NASA's James Webb Space Telescope has safely arrived at its launch site in French Guiana, on the northeastern coast of South America, technical teams have begun making progress on the final checklist of preparations before liftoff later this year.

These preparations are expected to last 55 days from the observatory's arrival by ship to the day of launch.

After Webb arrived at the Arianespace clean room facilities in French Guiana, contamination control technicians ensured the observatory is clean and contaminant free following its 5,800 mile journey. Then engineers ran a final set of electrical and functional tests and checked the stowed mechanical configuration to ensure delivery went smoothly. A trained crew in special hazmat suits will soon begin the two-week process of loading the spacecraft with the hydrazine fuel and nitrogen tetroxide oxidizer it will need to power its rocket thrusters to maintain its orbit. Next, Webb will move to the nearby vehicle integration building to be lifted and mounted on top of the Ariane 5 rocket "stack." The final few remove-before-flight "red-tag" items are taken off, and a few remaining add-before-flight "green tag" items are installed. Then the rocket fairing is lifted and lowered over top and locked into place, signifying the conclusion of a long journey. At this point, Webb will be very nearly ready to launch from Europe's Spaceport, also known as the Guiana Space Center (CSG).



Upon its arrival at the launch site in Kourou, French Guiana, engineers quickly set about unpacking, cleaning, and preparing the James Webb Space Telescope in its remaining days on Earth.

Credits: NASA/Chris Gunn

As a fully integrated launch vehicle with Webb as the payload, the Ariane rocket will roll out to the launch pad a few days before launch. Engineers monitor the rocket via electrical connections running from the payload control room to the pad through an umbilical attachment to the vehicle that separates at liftoff. A few hours before liftoff, the rocket is loaded with liquid hydrogen fuel and liquid oxygen oxidizer. About a half hour before launch, engineers in the payload control room switch the spacecraft from external electrical power to the spacecraft's on-board battery.

Webb's launch will be a pivotal moment for NASA and its partners, ESA (European Space Agency) and the Canadian Space Agency (CSA), but it is only the beginning. The following 29 days will be an exciting but harrowing time. Thousands of parts must work correctly, in sequence, to unfold Webb and put it in its final configuration, all while it flies through the expanse of space alone, to a destination nearly one million miles away.

A more detailed breakdown of what lies ahead for Webb:

Webb's 29 days on the edge begin upon liftoff. After 206 seconds of flight, at an altitude of about 75 miles above the atmosphere, the two halves of the rocket fairing that shields the observatory during ascent are separated by a pyrotechnic system with springs that expose the observatory to space. Ground teams expect to receive communication from Webb shortly after separation. Webb will then separate from the launch vehicle nearly 28 minutes after launch, and from this point on the ground team at the Space Telescope Science Institute in Baltimore will be in full control, to begin the most complex sequence of deployments ever attempted in a single space mission.

To unfold the history of the universe, we must first unfold this telescope. Following launch, over 300 single point failure items and 50 major deployments must work to ensure optimal science.

Webb's first deployment, the extension of its solar array, will occur between 31 to 33 minutes after liftoff, stopping the drain on the observatory's internal battery by supplying nearly 2 kilowatts of power to drive the spacecraft's electrical systems and avionics. To enable the highest data rate communication to the ground through NASA's Deep Space Network (DSN), the onboard medium and high-gain antenna platform is deployed at two hours.

Continued on Page 12



At 12 and a half hours after launch, Webb will fire its thrusters, performing the first of several critical course corrections that send the observatory towards its final destination in orbit. The observatory will pass the Moon nearly two and a half days after launch, faster than the time it took Apollo astronauts to reach lunar orbit.

Webb's first large deployment, the extension of its sunshield frame known as a unitized pallet structure, folds down nearly three days after launch, opening the observatory up to continue expanding. This represents the start of all major deployments and is scheduled to take approximately five hours for both front and back pallets to fold down completely.

Four days after launch, a deployable tower will extend to separate the telescope mirrors and instruments from the spacecraft bus. This separation effectively isolates the telescope from vibrations and conducted heat coming from the spacecraft bus. Additionally, this extension allows for the rest of Webb's larger deployable components, like its sunshield and primary mirror, to have enough room to make their own sequence of complex movements afterwards.

Sunshield membrane deployments formally begin approximately five days after launch, as special covers that protect the sunshield during ascent will roll out of the way. Next, a critical juncture in the mission will occur when all of the 107 sunshield release mechanisms, or special pins that keep the five sunshield layers locked into place, need to fire on cue and pull themselves out to free the membranes. After all sunshield pins have been successfully removed, two wings, known as mid-booms, extend to pull each of the sunshield layers out into their characteristic diamond formation nearly a day later. Following full deployment, each of the five layers are tensioned and separated using special pulleys and motor systems. Sunshield deployments and tensioning are expected to conclude between eight to nine days after liftoff but can be slowed down to circumvent any unforeseen issues if they arise.

Following the conclusion of sunshield tensioning, a special radiator behind the primary mirror is deployed to help cool down the scientific instruments.

Next, Webb's optics, and NASA's new eye on the cosmos, open up. **Telescope deployment** begins by unfolding and latching into place the tripod holding the **secondary mirror**, and it is expected to conclude two hours into the 10th day after liftoff. The secondary mirror is one of the most important pieces of equipment on the telescope, essential to the success of the mission. This smaller circular mirror plays an important role in collecting light from Webb's 18 primary mirrors into a focused beam. **Primary mirror** deployment is set to begin on the 12th day, with the mirror's side panels, each holding three primary mirror segments, taking nearly three hours to extend out and latch into place. At 13 days in, Webb's large-scale deployments are expected to conclude with the locking in of its primary mirror wings, revealing the telescope in all its glory.

A 10-day, multi-step process to move all 18 primary mirror segments out of their launch configuration will begin after the mirror wings are latched in and conclude on day 25. To begin fine-tuning the mirrors, 126 extremely precise actuators on the backside of the mirrors will position and subtly bend or flex each mirror into a specific prescription, a process that will take months.

On the 29th day, Webb will fire its thrusters once again to insert itself into its prescribed orbit at the second Lagrange point, or L2, nearly one million miles away from Earth, formally concluding the most difficult and complex deployment sequence ever attempted in space.



NASA's James Webb Space Telescope seen being prepared for transport to French Guiana for launch later this year.

Credits: NASA/Chris Gunn

The James Webb Space Telescope will be the world's premier space science observatory when it launches in 2021. Webb will solve mysteries in our solar system, look beyond to distant worlds around other stars, and probe the mysterious structures and origins of our universe and our place in it. Webb is an international program led by NASA with its partners, ESA (European Space Agency), and the Canadian Space Agency



NASA's James Webb Space Telescope rolls to its final stop before launch from ArianeSpace's ELA-3 launch complex at Europe's Spaceport in French Guiana, on the northeastern coast of South America.

Credit: NASA/Chris Gunn





Kid Spot Jokes:

Q: Why does the moon orbit the Earth?

A: To get to the other side??

Q: How do astronomers organize a party?

A: They planet.

Q: What music do Astronauts listen to?

A: Nep-Tunes!

Snoopy to Fly on NASA's Artemis 1 Moon Mission

Credit: NASA

For more than 50 years, Snoopy has contributed to the excitement for NASA human spaceflight missions, helping inspire generations to dream big. NASA has shared an association with Charles M. Schulz and Snoopy since Apollo missions and continues under Artemis with new educational activities. Up next -- Snoopy will ride along as the zero gravity indicator on Artemis I.

Artemis I is an uncrewed flight test of the Space Launch System (SLS) rocket and Orion spacecraft around the Moon launching in early 2022 before missions with astronauts. Zero gravity indicators are small items carried aboard spacecraft that provide a visual indicator when a spacecraft has reached the weightlessness of microgravity. Without astronauts aboard Orion, Snoopy will help share the journey with the world as he rides along in the cabin with a manikin and two other "passengers."

Apollo Era

Already a well-known household character, Snoopy was used to encourage NASA's spaceflight safety initiative during the time of Apollo. Schulz created comic strips of Snoopy on the Moon, capturing public excitement about America's achievements in space.

Already a well-known household character, Snoopy was used to encourage NASA's spaceflight safety initiative during the time of Apollo. Schulz created comic strips of Snoopy on the Moon, capturing public excitement about America's achievements in space.

The agency's Silver Snoopy award was created during the Apollo era and remains to this day. It is a high honor awarded to NASA employees and contractors by astronauts, celebrating achievements related to mission success and human flight safety. Each silver pin given with this award, depicting astronaut Snoopy, was flown in space. Continuing the tradition, Artemis I will also carry a package of silver snoopy pins for future recognitions.

Today, the partnership continues. For Snoopy's flight on the Artemis I mission, he will be outfitted in a custom orange flight suit complete with gloves, boots, and a NASA patch. Peanuts is releasing a new suite of curriculum and

short videos with its partner, GoNoodle, to encourage kids to learn about gravity, teamwork, and space exploration while they follow Snoopy along on his Artemis I journey .



Snoopy as a zero gravity indicator.

Credits: 2021 Peanuts Worldwide LLC



From the Board of Directors

Announcements

- ◇ San Jose Astronomical Association is closely monitoring the coronavirus/COVID-19 situation in Santa Clara County. In accordance with guidance from Santa Clara County Health Department, SJAA leaders have been canceling many events, especially those that involve close contact and shared surfaces on telescopes. Some events, however, have successfully been moved to an online format. Please check our Meetup to determine if an event has been canceled or moved online. The board of directors will periodically review the status of the outbreak and the guidance from the county to determine the impact upon upcoming events.
- ◇ New board member elections will be held at the February 2022 board meeting on February 19. The following 4 board positions will be open for Elections: Director 2, Director 4, Director 6, and Director 8. All current incumbents are expected to run for their positions again. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details.
- ◇ New officer elections for President, Vice-President, Treasurer and Secretary positions will be held at the March 2022 board meeting on March 19. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details.

Board Meeting Excerpts

Sept 18, 2021

In attendance

Swami, Rob J, Glenn, Wolf, Vini, Peter, Sukhada,
Absences: Ken, Sandy
Guests: Leslie

Donations

SJAA has received a number of equipment donations over the past few months. Sales of items have resulted in above \$200+ being added to the general fund.

Alternate/Backup Home for SJAA

The task force discussed the activities for which SJAA uses Houge Park, and brainstormed to develop a list of potential alternate sites in weekly meetings.

Insurance authorization

The Board approved a motion, moved by Rob and seconded by Swami, to authorize the SJAA treasurer (Rob) to spend up to \$650 per year to purchase the required insurance, with the caveat that this insurance is to be effective no earlier than on the date that the City allows SJAA full access to Building 1 at Houge Park

"Time Box" for Membership Report

By a vote of five (in favor), one (opposed) and one (abstain), the Board voted to move from the Lunar calendar to the Gregorian Calendar for reporting membership

October 23, 2021

In attendance

Swami, Sukhada, Wolf, Vini, Glenn Ken,, Peter
Absences: Sandy, Rob J
Guest: Leslie, Kanch

Alternate/Backup Home for SJAA

Swami met with the Evergreen Valley College (EVC)/Montgomery Hill Observatory (MHO) staff who were quite motivated to work with SJAA. They also want student involvement in the SJAA activities on campus. Ideas were brainstormed on how the two organizations could collaborate on activities and events.

The Board expressed interest in moving ahead with this possibility while continuing to flesh out the other possibilities. The task force will continue its work and report back to the Board.

Ongoing relationship SJAA and the City of San Jose

Swami reported that he met with the Arab Community Center and the Youth Shakespeare Group who also use Houge Park. A meeting with the City is planned in early November after finalizing an agreement proposal that's to be submitted to the City.

November 20, 2021

In attendance

Ken, Swami, Wolf, Vini, Sukhada, Peter,
Absences: Glenn, Sandy
Guests: Kanch, Leslie

Alternate/Backup Home for SJAA

Swami reported that Evergreen Valley College continues to be enthusiastic about partnering with SJAA. However, he cautioned that the bureaucratic hurdles within EVC "will take time" to overcome. Dave Ittner was shown Tilden Ranch by Terry at Open Space Authority. Board members wish to hear more about this from Dave.

Website Monitoring

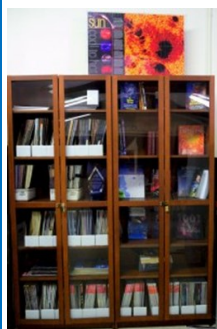
Webcity Media will end its monitoring, updating and backup services after May 2022. Rob will contact our webmaster, Satish. He suggested that a task force be created to perhaps redesign the SJAA site. Swami wondered if the webmaster could chair the task force, and Rob agreed to ask him.

Open Leadership Positions

Peter recommended a Board-led group to contact other members and encourage leadership building. Wolf suggested a pizza night to encourage members to get reacquainted. Both Peter and Swami suggested an 'open house' outdoors. The board will alert membership to the need for volunteers.



SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held quarterly. The next Quick Start session during this time period is October 12th.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content chang-

ing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Ken Miura
Vice President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Peter Melhus
Director:	Sandy Mohan
Director:	Vini Carter
Director:	Swami Nigam
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Abishek
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Hy Murveit
Intro to the Night Sky:	David Grover
In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal
Publicity:	Beth Johnson
Quick START	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Vacant
Refreshments:	Marianne Damon
Social Manager:	Beth Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
E-mails:	http://www.sjaa.net/contact

SJAA *Ephemeris*, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
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San Jose Astronomical Association Annual Membership Form

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Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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